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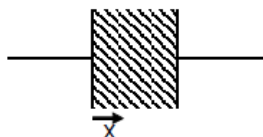
JEE Mains 2020 Jan Chapter wise Question Bank

Capacitance

Q1

A parallel plate capacitor with plate area A & plate separation d is filled with a dielectric material of dielectric constant given by $k = k_0(1 + \alpha x)$. Calculate capacitance of system: (given $\alpha d \ll 1$)

एक समान्तर पट्ट संधारित्र जिसकी सतह का क्षेत्रफल A है तथा पट्टिकाओं के मध्य की दूरी d है। इनके मध्य एक परावैद्युतांक पदार्थ भरा है। पदार्थ का परावैद्युतांक नियतांक $k = k_0(1 + \alpha x)$ है। तब निकाय की धारिता होगी: (दिया है $\alpha d \ll 1$)



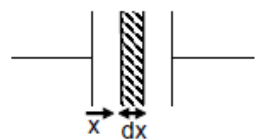
- (1) $\frac{k_0 \epsilon_0 A}{d} (1 + \alpha^2 d^2)$ (2) $\frac{k_0 \epsilon_0 A}{d} \left(1 + \frac{\alpha d}{2}\right)$ (3) $\frac{k_0 \epsilon_0 A}{2d} (1 + \alpha d)$ (4) $\frac{k_0 \epsilon_0 A}{2d} \left(1 + \frac{\alpha d}{2}\right)$

7th Jan Morning

Sol

(2)

Capacitance of element घटक की धारिता = $\frac{k \epsilon_0 A}{dx}$



Capacitance of element घटक की धारिता, $C' = \frac{k_0(1 + \alpha x) \epsilon_0 A}{dx}$

$$\sum \frac{1}{C'} = \int_0^d \frac{dx}{k_0 \epsilon_0 A (1 + \alpha x)}$$

$$\frac{1}{C} = \frac{1}{k_0 \epsilon_0 A \alpha} \ln(1 + \alpha d)$$

Given दिया है— $\alpha d \ll 1$

$$\frac{1}{C} = \frac{1}{k_0 \epsilon_0 A \alpha} \left(\alpha d - \frac{\alpha^2 d^2}{2} \right)$$

$$\frac{1}{C} = \frac{d}{k_0 \epsilon_0 A} \left(1 - \frac{\alpha d}{2} \right)$$

$$C = \frac{k_0 \epsilon_0 A}{d} \left(1 + \frac{\alpha d}{2} \right)$$

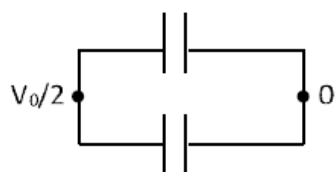
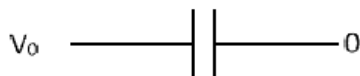
Q2

A capacitor of 60 pF charged to 20 volt. Now battery is removed and then this capacitor is connected to another identical uncharged capacitor. Find heat loss in nJ.

7th Jan Evening

Sol

6



$$V_0 = 20 \text{ V}$$

Heat loss ऊष्मा हानि = $U_i - U_f$

$$\begin{aligned}
 &= \frac{1}{2} C V_0^2 - 2 \left[\frac{1}{2} C \left(\frac{V_0}{2} \right)^2 \right] \\
 &= \frac{C V_0^2}{4} \\
 &= \frac{(60 \times 10^{-12})(20)^2}{4} \text{ J} \\
 &= 6 \times 10^{-9} \text{ J} = 6 \text{ nJ}
 \end{aligned}$$

Q3

If two capacitors C_1 & C_2 are connected in parallel then equivalent capacitance is $10\mu\text{F}$. If both capacitor are connected across 1V battery then energy stored by C_2 is 4 times that of C_1 . Then the equivalent capacitance if they are connected in series is –

दो संधारित्र जिनकी धारिताएँ C_1 & C_2 हैं को समान्तर क्रम में जोड़ने पर प्राप्त तुल्य धारिता $10\mu\text{F}$ है। यदि दोनों संधारित्र को 1V की बैटरी के साथ जोड़ा जाता है तो C_2 में संग्रहित ऊर्जा, C_1 में संग्रहित ऊर्जा की 4 गुना है। संधारित्रों को श्रेणी क्रम में जोड़ने पर इनकी तुल्य धारिता क्या होगी।

(1) $1.6\mu\text{F}$

(2) $16\mu\text{F}$

(3) $4\mu\text{F}$

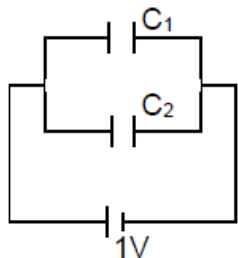
(4) $\frac{1}{4}\mu\text{F}$

8th Jan Morning

Sol

Capacitance
(1)

JEE Mains 2020 Jan Chapter wise Question Bank



Given दिया है $C_1 + C_2 = 10\mu\text{F}$... (i)

$$4\left(\frac{1}{2}C_1V^2\right) = \frac{1}{2}C_2V^2$$

$$\Rightarrow 4C_1 = C_2 \quad \dots (ii)$$

from equation (i) & (ii) समी. (i) तथा (ii) से

$$C_1 = 2\mu\text{F}$$

$$C_2 = 8\mu\text{F}$$

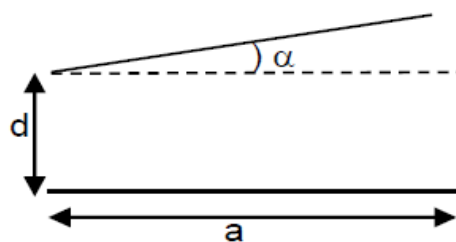
If they are in series यदि इनको श्रेणी क्रम में जोड़ा जाए

$$C_{eq.} = \frac{C_1C_2}{C_1 + C_2} = 1.6\mu\text{F}$$

Q4

Two square plates of side 'a' are arranged as shown in the figure. The minimum separation between plates is 'd' and one of the plate is inclined at small angle α with plane parallel to another plate. The capacitance of capacitor is (given α is very small)

दिये गये चित्र के अनुसार भुजा 'a' की दो वर्गाकार प्लेटे व्यवस्थित की गई है। प्लेटों के बीच की न्यूनतम दूरी d है और एक प्लेट दुसरी प्लेट के समान्तर रेखा से छोटे कोण α पर झुकी है इस संधारित्र की धारिता होगी।



(1) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{2d}\right)$

(2) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{d}\right)$

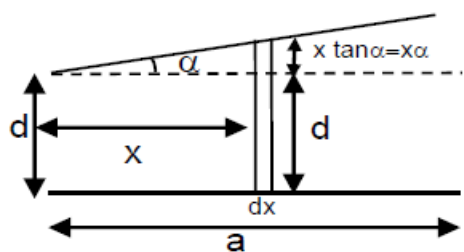
(3) $\frac{\epsilon_0 a^2}{d} \left(1 + \frac{\alpha a}{2d}\right)$

(4) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{4d}\right)$

8th Jan Evening

Sol

1



$$dc = \frac{\epsilon_0 a dx}{d + \alpha x}$$

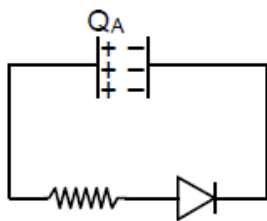
$$\Rightarrow c = \frac{\epsilon_0 a}{\alpha} [\ln(d + \alpha x)]_0^a$$

$$= \frac{\epsilon_0 a}{\alpha} \ln\left(1 + \frac{\alpha a}{d}\right) \approx \frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{2d}\right)$$

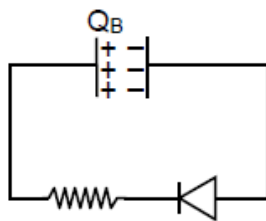
Q6

A Capacitor C and resistor R are connected to a battery of 5V in series. Now battery is disconnected and a diode is connected as shown in figure (a) and (b) respectively. Then charge on the capacitor after time RC in (a) and (b) respectively is Q_A and Q_B . Their value are

एक संधारित्र C तथा प्रतिरोध R, 5V की बैटरी के साथ श्रेणी क्रम जुड़ा है। अब बैटरी को हटा दिया जाता है तथा एक डायोड क्रमशः चित्र (a) तथा (b) के अनुसार जोड़ा जाता है। यदि समय RC के पश्चात् संधारित्र पर आवेश (a) तथा (b) में क्रमशः Q_A तथा Q_B है तो इनके मान ज्ञात करो—



(a)



(b)

(1) 5CV, 5CV

(2) $\frac{5CV}{e}, \frac{5CV}{e}$

(3) $\frac{5CV}{e}, 5CV$

(4) 5CV, $\frac{5CV}{e}$

9th Jan Evening

Sol

Capacitance

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(3)

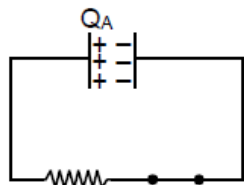
Maximum charge on capacitor संधारित्र पर अधिकतम आवेश = $5CV$

(a) is forward biased and (b) is reverse biased

for case (a)

(a) अग्रबायस है तथा (b) पश्चबायस है।

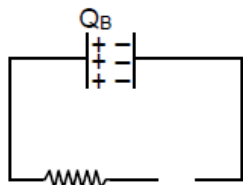
(a) के लिये



$$\text{so } q = q_{\max} [e^{-t/RC}]$$

$$= 5 CV$$

$$Q_A = 5CVe^{-1}$$



$$Q_B = 5CV$$



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